

Endangered Species Analysis in National Parks

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Project Overview

- **Goal:** Analyze conservation statuses and observation patterns of species in national parks.
- **Datasets:**
 - `species_info.csv`: Species details (category, scientific name, conservation status).
 - `observations.csv`: Sightings over 7 days in 4 parks.
- **Objectives:**
 - Distribution of conservation statuses.
 - Identify high-risk categories.
 - Examine observation trends.

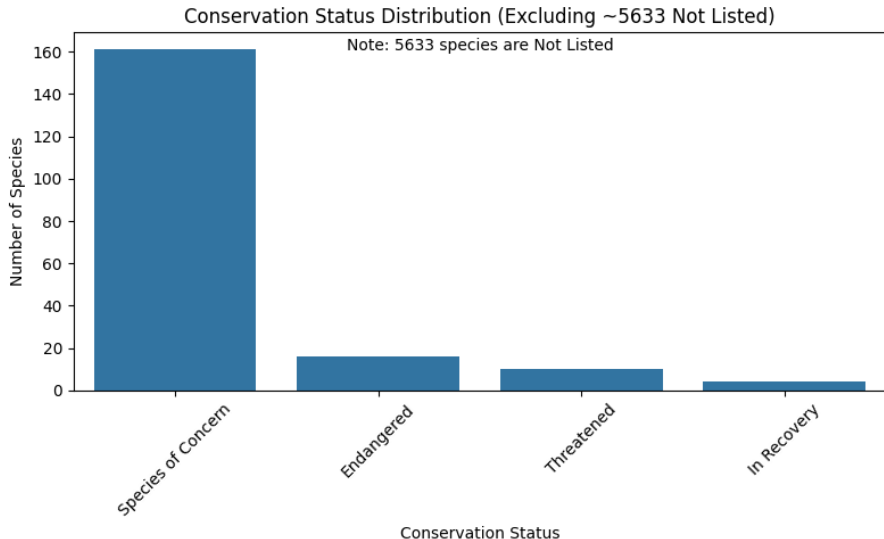
- **species_info.csv:**

- 5824 species, 5541 unique scientific names.
- Columns: `category`, `scientific_name`, `common_names`, `conservation_status`.
- Cleaning: Imputed 5633 missing `conservation_status` as "Not Listed".

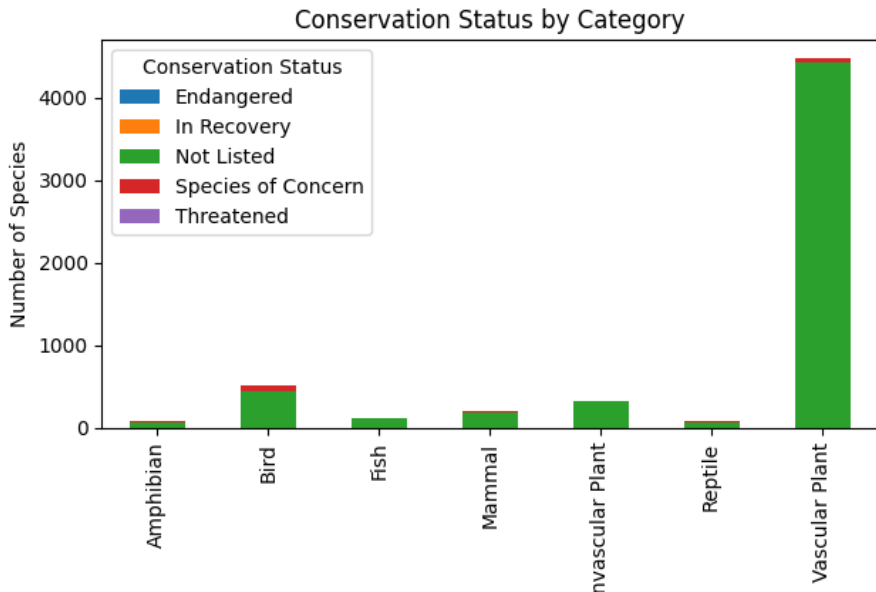
- **observations.csv:**

- 23281 sightings (after removing 15 duplicates).
- Columns: `scientific_name`, `park_name`, `observations`.
- 4 parks: Great Smoky Mountains, Yosemite, Bryce, Yellowstone.

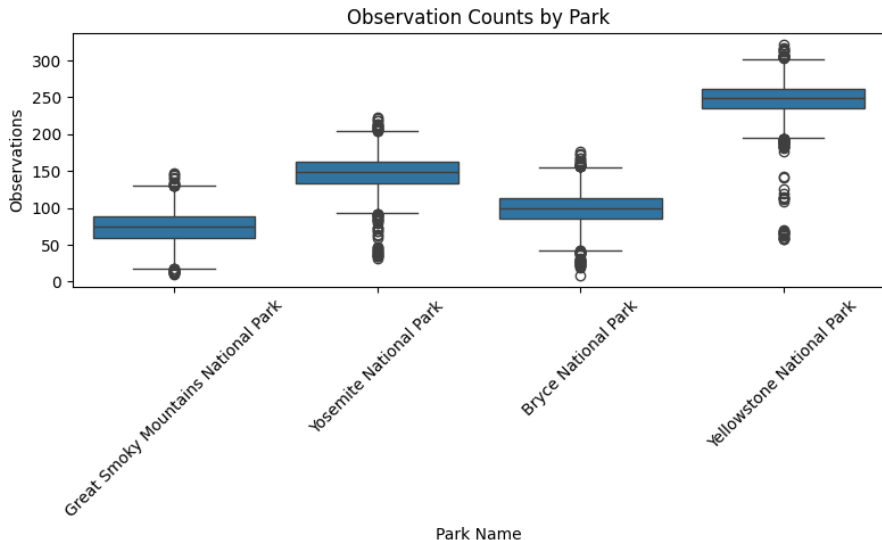
Visualization: Conservation Status



Visualization: Status by Category



Visualization: Observations by Park



Key Findings

- **Conservation Status:** 5633/5824 species are "Not Listed"; 191 are Species of Concern (161), Endangered (16), Threatened (10).
- **Category Patterns:** Vascular Plants (4470) dominate, followed by Birds (521), Mammals (214). Mammals/Birds have higher Endangered/Threatened proportions.
- **Observations:** Yellowstone has highest observations (1.4M), suggesting greater survey effort or diversity. Average observations 100–150 per species.

Conclusions and Next Steps

- **Conclusions:**

- Mammals and Birds are key conservation priorities due to higher endangerment.
- Yellowstone's high observations warrant further study.

- **Next Steps:**

- Analyze observation counts for endangered species (rarity).
- Investigate park-specific conservation needs.
- Expand analysis to temporal trends or additional parks.